

Five-number summary

Synthetic data. A five-number summary gives minimum, Q1, median, Q3, and maximum.

What is the summary for 2, 4, 4, 6, 8, 10, 10, 12?

Box plot terminology

A box plot, box-and-whisker plot, or box-whisker plot displays quartiles with a box and whiskers.

Which values mark the box edges and its median line?

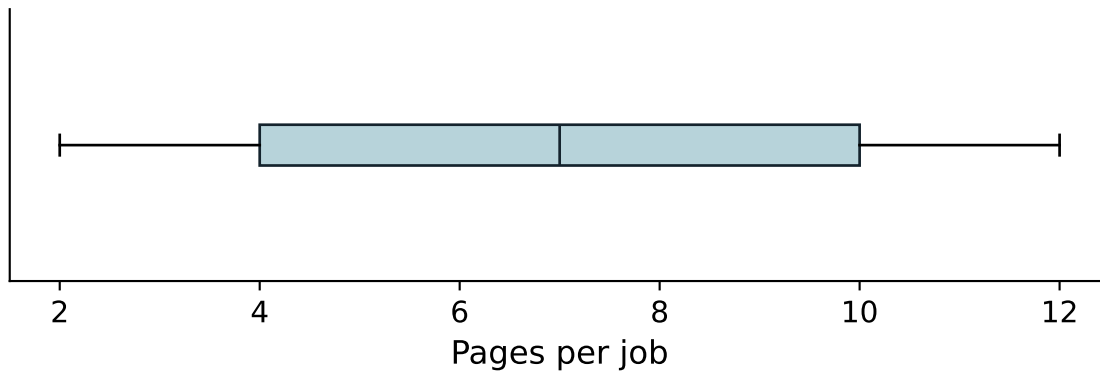
Build a box plot

Synthetic pages per job: minimum 2, $Q1 = 4$, median 7, $Q3 = 10$, maximum 12.

How would you draw a box plot on a scaled number line?

Read the median

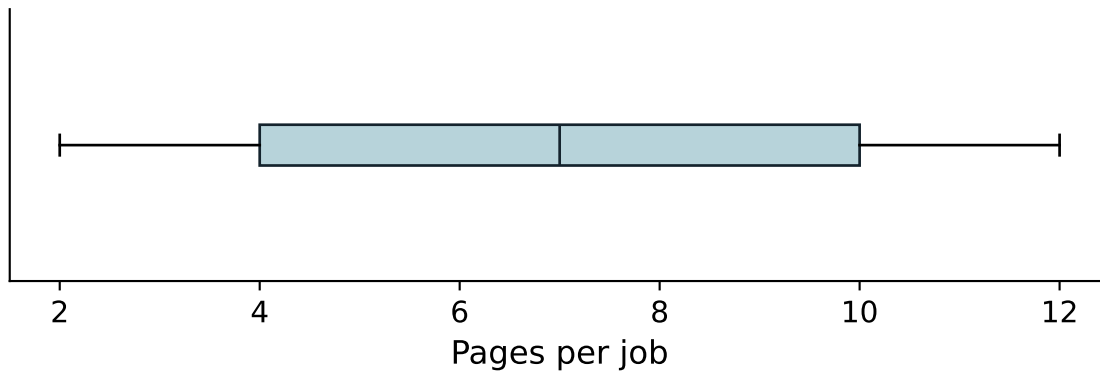
Synthetic pages; min/max whiskers.



What is the median?

Read the box

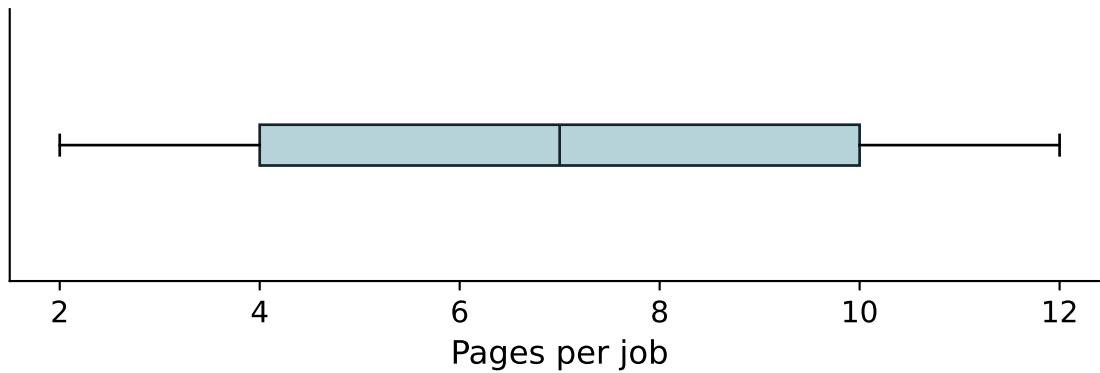
Synthetic pages; min/max whiskers.



What interval contains approximately the middle half?

Read the range

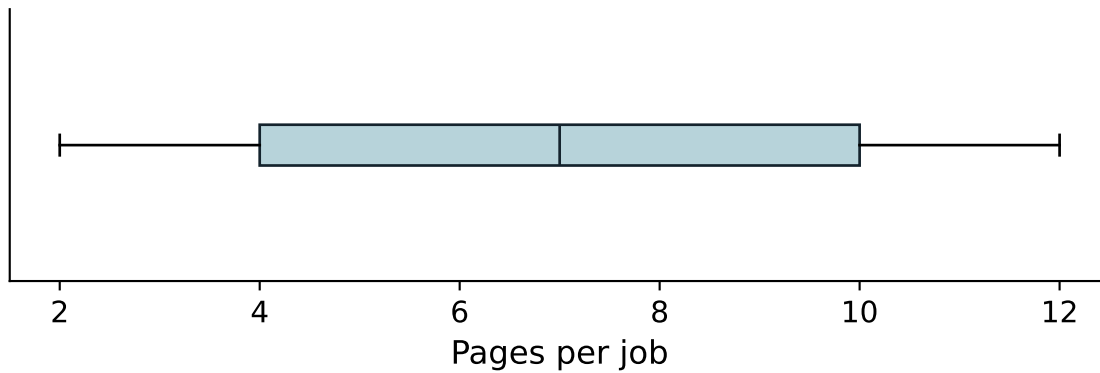
Synthetic pages; min/max whiskers.



What is the range?

Read the IQR

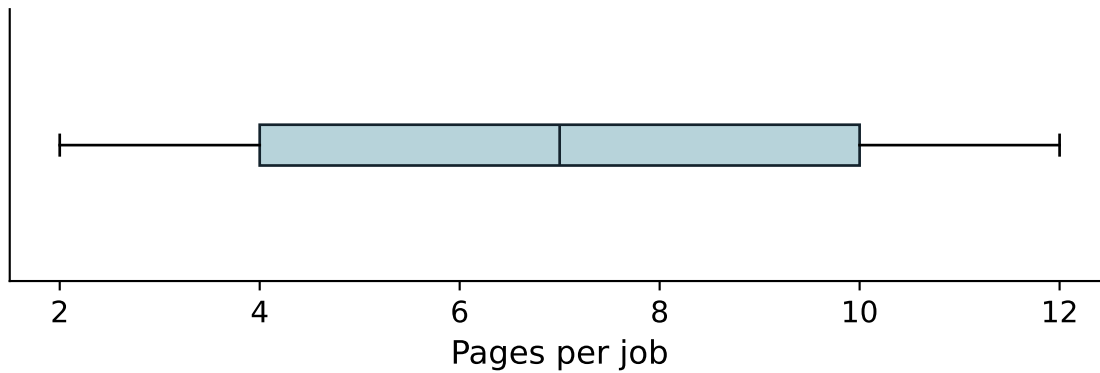
Synthetic pages; min/max whiskers.



What is the IQR?

Length is not count

Synthetic pages; min/max whiskers.



Does a longer quarter of the plot necessarily contain more observations?

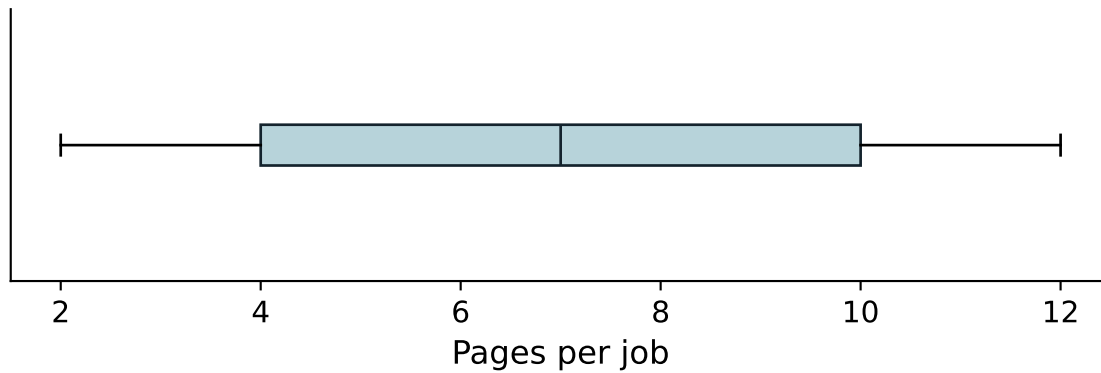
Coinciding quartiles

Synthetic five-number summary: 1, 1, 5, 5, 7.

How would a box plot show the minimum equaling Q1 and the median equaling Q3?

Within a quarter

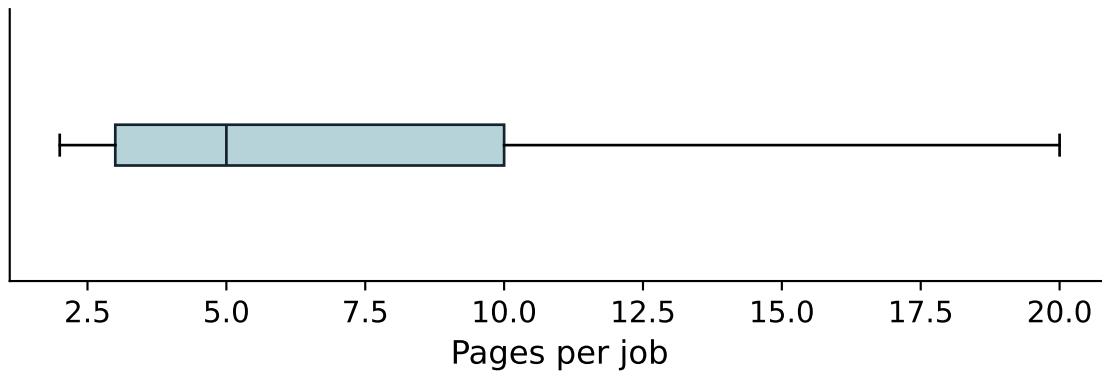
Synthetic pages; min/max whiskers.



Can the plot determine exactly how many jobs printed 5 pages?

Concentration

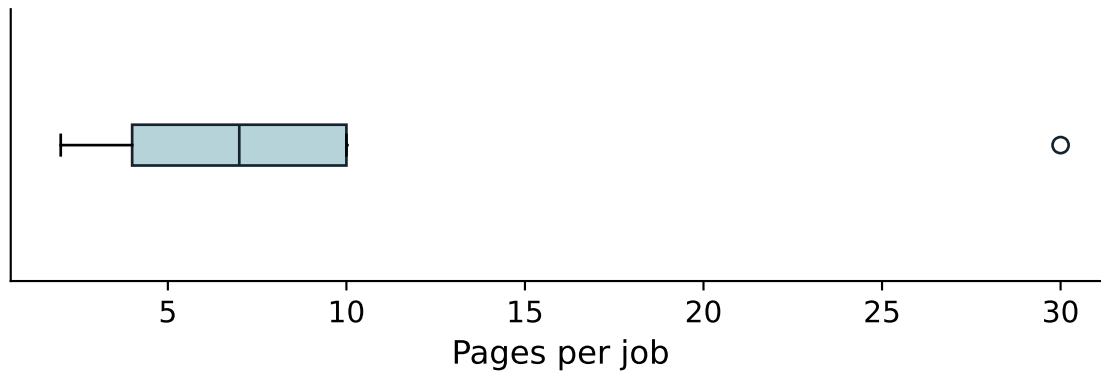
Synthetic pages; min/max whiskers.



Which quarter spans the shortest interval?

A modified box plot

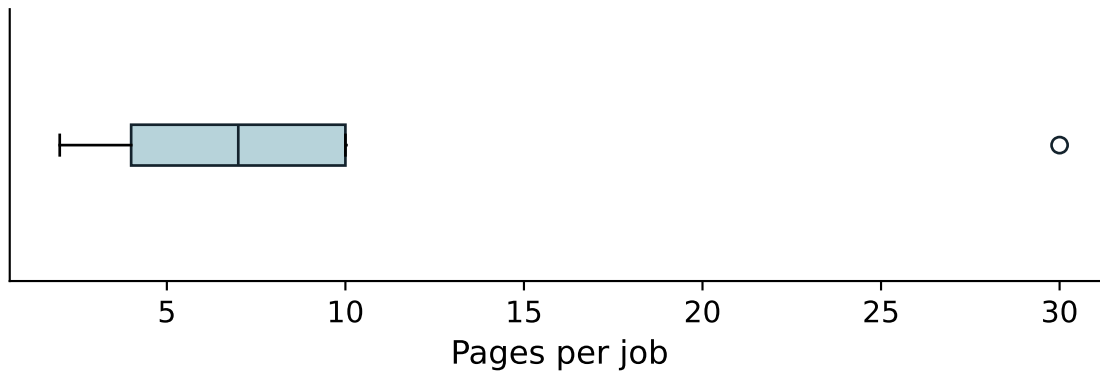
Synthetic pages; outliers are separate points.



What does the point at 30 pages represent?

The whisker endpoint

Synthetic pages; outliers are separate points.



Why does the upper whisker stop before 30?

Compare centers

Synthetic five-number summaries. A: 2, 4, 7, 10, 12. B: 5, 7, 10, 13, 15.

Which group has the higher median?

Compare middle spreads

Synthetic five-number summaries. A: 2, 4, 7, 10, 12. B: 2, 6, 7, 8, 12.

Which group has the larger IQR?

Same summary, different data

Two synthetic datasets have the same five-number summary.

Must their individual observations be identical?

Match the graph to a question

We want to compare the middle half of file sizes for two servers.

What makes a box plot useful here?

Investigate an outlier

A modified box plot flags one large print job.

What additional information would help interpret that job?